AlphaPulse – Portfolio Risk & Volatility Monitor

FINANCIAL ANALYTICS - PORTFOLIO RISK & VOLATILITY MONITOR

Project Report: Towards the fulfillment of the project under three months internship as Data Analytics Associate (L1) in Infotact Solutions

Project Title: AlphaPulse – Portfolio Risk & Volatility Monitor

Project Type: Financial Analytics & Risk Modeling System

Technologies Used: Python · SQL (SQLite) · Power BI · yfinance · Pandas · NumPy

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CHAPTER 1 — EXECUTIVE SUMMARY

AlphaPulse is an end-to-end financial analytics and risk modeling system built to analyze portfolio performance, measure market risk, and simulate future investment outcomes. The system integrates real-world stock market data from Yahoo Finance, processes it through a structured Python-based data pipeline, stores it in a relational SQLite database, and delivers actionable insights through an interactive Power BI dashboard.

The project covers five major technology companies — Apple (AAPL), Microsoft (MSFT), Alphabet (GOOG), Tesla (TSLA), and Amazon (AMZN) — and applies industry-standard quantitative techniques including daily log return computation, rolling volatility analysis, correlation matrix construction, Monte Carlo simulation with over 10,000 portfolio scenarios, and Value at Risk (VaR) estimation at the 95% confidence level.

The final deliverable is a business-ready dashboard that consolidates all risk metrics, trend analyses, and simulation results into a single, interpretable visual interface — enabling data-driven portfolio management decisions.

CHAPTER 2 — INTRODUCTION

2.1 Background

Financial markets are inherently volatile and uncertain. For investors, portfolio managers, and financial analysts, understanding the risk embedded in a portfolio of assets is as important as understanding its returns. The discipline of quantitative finance provides a set of mathematical and statistical tools to measure, monitor, and manage this risk systematically.

Traditionally, such analysis was confined to large financial institutions that could afford expensive enterprise platforms like Bloomberg Terminal or FactSet. With the emergence of open-source Python libraries, free financial APIs, and accessible business intelligence tools, it is now possible to build professional-grade financial analytics systems independently.

2.2 Project Context

AlphaPulse was developed as a practical demonstration of how data engineering, statistical modeling, and business intelligence can be unified into a single analytical system. The project simulates the kind of portfolio risk monitoring system used by asset management firms, hedge funds, and financial risk teams.

2.3 Motivation

The key motivation behind this project was to bridge the gap between raw financial data and decision-ready risk intelligence — making sophisticated financial analysis accessible, reproducible, and visually intuitive.

CHAPTER 3 — PROBLEM STATEMENT

Portfolio risk management in modern financial markets presents several persistent challenges:

3.1 Lack of Real-Time Risk Visibility Asset managers and retail investors often rely on static reports or lagging indicators. There is no consolidated, real-time view of how risk is evolving across their holdings.

3.2 Complexity of Multi-Asset Volatility Tracking Tracking volatility across multiple assets simultaneously — and understanding how individual asset volatility contributes to overall portfolio risk — requires continuous recalculation that manual methods cannot sustain.

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3.3 Limited Diversification Insight Most basic tools do not quantify the correlation between assets. Without understanding inter-asset correlation, investors cannot make informed diversification decisions or understand whether their portfolio is genuinely diversified.

3.4 Absence of Forward-Looking Risk Estimation Historical analysis alone is insufficient. Investors need simulation-based forward-looking tools to understand what their portfolio could look like under a range of future market conditions.

3.5 Inaccessibility of Quantitative Risk Metrics Metrics like Value at Risk (VaR), which are standard in institutional finance, are rarely accessible to individual analysts or smaller firms without enterprise-grade software.

AlphaPulse directly addresses each of these challenges through a unified, open-source, data-driven system.

CHAPTER 4 — PROJECT OBJECTIVES

The following objectives guided the design and development of AlphaPulse:

1. **Data Acquisition:** Programmatically fetch real-time and historical OHLCV (Open, High, Low, Close, Volume) stock data for a defined portfolio of assets using the Yahoo Finance API.
2. **Data Engineering:** Build a robust data cleaning and transformation pipeline that ensures time-series continuity, handles missing values, and stores structured data in a relational database.
3. **Performance Analysis:** Compute daily log returns and analyze individual asset performance over the selected time period.
4. **Risk Measurement:** Calculate rolling volatility (30-day standard deviation of log returns) and construct a correlation matrix to quantify inter-asset relationships.
5. **Portfolio Modeling:** Aggregate individual asset metrics into a portfolio-level view, computing weighted portfolio returns and overall portfolio volatility accounting for cross-asset correlations.
6. **Scenario Simulation:** Implement Monte Carlo simulation to generate 10,000+ randomized portfolio weight scenarios and map the risk-return distribution across all simulated portfolios.
7. **Downside Risk Quantification:** Estimate Value at Risk (VaR) at the 95% confidence level to provide a clear, single-number measure of potential downside exposure.
8. **Dashboard Delivery:** Design and publish an interactive Power BI dashboard that consolidates all analytical outputs into a business-ready visual interface.

CHAPTER 5 — SCOPE OF THE PROJECT

5.1 In Scope

- Historical and real-time data acquisition for 5 equity assets (AAPL, MSFT, GOOG, TSLA, AMZN)
- Python-based data processing and quantitative modeling pipeline
- SQLite-based structured data storage
- Rolling volatility, correlation, portfolio risk, Monte Carlo simulation, and VaR analysis
- Interactive Power BI dashboard with 5 analytical views

5.2 Out of Scope

- Live algorithmic trading or order execution
- Options, derivatives, or fixed income instruments
- Machine learning-based price prediction models

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- Multi-currency or international exchange data
- User authentication or cloud deployment

5.3 Target Users

- Financial analysts and portfolio managers
- Data analysts working in fintech or investment domains
- Academic researchers studying quantitative finance
- Data science students building domain-specific projects

CHAPTER 6 — SYSTEM ARCHITECTURE

6.1 Architecture Overview

AlphaPulse follows a layered, pipeline-driven architecture with four distinct stages:

```
[ Data Source Layer ]
      ↓
Yahoo Finance API (via yfinance)
      ↓
[ Data Processing Layer ]
      ↓
Python Pipeline → Cleaning → Transformation → Feature Engineering
      ↓
[ Storage Layer ]
      ↓
SQLite Relational Database
      ↓
[ Analytics & Visualization Layer ]
      ↓
Power BI Interactive Dashboard
```

6.2 Layer Descriptions

1. **Layer 1 — Data Source:** Raw OHLCV stock data is fetched from Yahoo Finance using the yfinance Python library. Data is pulled for all 5 assets over a defined historical period (typically 1–3 years).
2. **Layer 2 — Data Processing:** Python scripts handle data ingestion, cleaning (null handling, outlier removal, date alignment), transformation (wide to long format conversion), and computation of all financial metrics including log returns, rolling volatility, and correlation coefficients.
3. **Layer 3 — Storage:** Cleaned and transformed data is stored in an SQLite database with separate tables for raw prices, computed returns, volatility, and portfolio-level metrics. This enables structured querying and efficient data retrieval by Power BI.
4. **Layer 4 — Visualization:** Power BI connects to the SQLite database and renders all analytical outputs across 5 interactive dashboard panels, including trend charts, heatmaps, simulation scatter plots, and KPI cards.

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CHAPTER 7 — TECH STACK & JUSTIFICATION

7.1 Technology Choices

Tool / Library	Category	Justification
Python 3.x	Core Language	Industry-standard for data science and financial modeling; rich ecosystem
yfinance	Data Acquisition	Free, reliable wrapper for Yahoo Finance API; supports OHLCV historical data
Pandas	Data Manipulation	Essential for time-series operations, reshaping, and tabular data handling
NumPy	Numerical Computing	Efficient matrix operations required for correlation, volatility, and simulation
SciPy	Statistical Analysis	Used for percentile computation in VaR estimation
SQLite	Database	Lightweight, serverless relational DB; ideal for local structured storage
Power BI	Visualization	Enterprise-grade BI tool; supports interactive dashboards with direct DB connectivity
Jupyter Notebook	Development Environment	Enables iterative development, inline visualization, and documented analysis

7.2 Why Not Alternatives?

- **PostgreSQL over SQLite:** Unnecessary for single-user local analysis; SQLite is sufficient and zero-configuration
- **Tableau over Power BI:** Power BI offers tighter Python/SQL integration and is widely used in financial services
- **Matplotlib over Power BI:** Python plotting is useful for exploratory analysis but lacks interactivity for a business dashboard

CHAPTER 8 — DATA ACQUISITION & PROCESSING

8.1 Data Source

Stock market data was sourced from **Yahoo Finance** via the `yfinance` Python library. Yahoo Finance provides freely accessible, adjusted historical price data for publicly listed equities.

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Assets Selected:

Ticker	Company	Sector
AAPL	Apple Inc.	Technology
MSFT	Microsoft Corporation	Technology
GOOG	Alphabet Inc.	Technology / Communication
TSLA	Tesla Inc.	Automotive / Clean Energy
AMZN	Amazon.com Inc.	E-Commerce / Cloud

- **Time Period:** January 2020 – December 2023 (approximately 3 years of daily data)
- **Data Fields Fetched:** Open, High, Low, Close, Adjusted Close, Volume

8.2 Data Cleaning

The raw dataset underwent the following cleaning procedures:

- **Missing Value Handling:** Forward-fill (ffill) applied to propagate the last valid observation across non-trading days; any remaining nulls removed
- **Date Alignment:** All 5 assets aligned to a common trading calendar to ensure synchronous time-series comparison
- **Outlier Handling:** Extreme single-day price spikes reviewed and verified against known corporate events (splits, dividends)
- **Adjusted Close Usage:** All return calculations used Adjusted Close prices to account for stock splits and dividend distributions accurately

8.3 Data Transformation

- Converted raw price data from **wide format** (one column per asset) to **long format** (one row per asset-date pair) to optimize SQL storage and Power BI querying
- Created additional derived columns: daily log returns, 30-day rolling standard deviation, cumulative returns

8.4 Data Storage Schema

Three primary tables stored in SQLite:

- `stock_prices` — Raw adjusted close prices (Date, Ticker, Adj_Close, Volume)
- `stock_returns` — Computed log returns per asset per date (Date, Ticker, Log_Return)
- `portfolio_metrics` — Portfolio-level aggregated metrics (Date, Portfolio_Return, Portfolio_Volatility)

CHAPTER 9 — QUANTITATIVE ANALYSIS & FINANCIAL MODELING

9.1 Daily Log Returns

Log returns are the standard measure of asset performance in quantitative finance. They are computed as:

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Formula: $r_t = \ln(P_t / P_{t-1})$

Where P_t is the adjusted closing price at time t and P_{t-1} is the previous day's price.

Why Log Returns:

- Time-additive — log returns across periods can simply be summed
- Normally distributed assumption holds better for log returns than simple returns
- Removes the effect of price scale, enabling fair cross-asset comparison

9.2 Rolling Volatility (30-Day)

Volatility measures the degree of variation in asset returns over time. A 30-day rolling standard deviation was computed for each asset:

Formula: $\sigma_t = \text{std}(r_{t-29} \ldots r_t)$

- High rolling volatility indicates elevated market risk and uncertainty
- TSLA consistently exhibited the highest rolling volatility across the portfolio
- MSFT and AAPL showed comparatively stable, lower volatility profiles

9.3 Correlation Matrix

The correlation matrix quantifies the linear relationship between every pair of asset return series:

Formula: $\rho_{A,B} = \text{Cov}(r_A, r_B) / (\sigma_A \times \sigma_B)$

Interpretation:

- Values close to +1: Assets move in the same direction — no diversification benefit
- Values close to 0: Assets move independently — moderate diversification
- Values close to -1: Assets move in opposite directions — maximum diversification benefit

Key Finding: MSFT and AAPL showed high positive correlation (both technology stocks), while TSLA showed relatively lower correlation with the others — contributing meaningful diversification to the portfolio.

9.4 Portfolio Return & Risk

Portfolio Return: $R_p = \Sigma (w_i \times r_i)$ Weighted sum of individual asset returns where w_i is the allocation weight of asset i.

Portfolio Volatility: $\sigma_p = \sqrt{(w^T \Sigma w)}$ Where Σ is the covariance matrix and w is the weight vector.

This formula captures cross-asset covariance — a key insight is that portfolio volatility is always lower than the weighted average of individual volatilities when assets are not perfectly correlated, mathematically proving the benefit of diversification.

CHAPTER 10 — ADVANCED MODELING: MONTE CARLO SIMULATION

10.1 Overview

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Monte Carlo Simulation is a computational technique that models the probability of different outcomes by running a large number of randomized experiments. In portfolio analysis, it is used to explore the full space of possible portfolio constructions and understand the risk-return tradeoff across all of them.

10.2 Implementation

- **Number of Simulations:** 10,000 random portfolio weight combinations generated
- **Weight Generation:** For each simulation, a set of 5 random weights was generated using a uniform distribution and normalized to sum to 1 (ensuring a fully invested portfolio with no leverage)
- **Per-Simulation Computation:** Each weight combination was used to compute expected annual portfolio return and expected annual portfolio volatility using historical return and covariance data
- **Sharpe Ratio Calculation:** For each simulation, the Sharpe Ratio was computed as:

`Sharpe Ratio = (R_p - R_f) / σ_p`

Where `R_f` is the risk-free rate (approximated using the U.S. 10-year Treasury yield).

10.3 Output — Efficient Frontier

The 10,000 simulation results were plotted as a scatter cloud on a **Risk vs. Return** axis, color-coded by Sharpe Ratio. This visualization produces the **Efficient Frontier** — a curve representing the set of portfolios that offer the maximum return for each level of risk.

Key Insights from Simulation:

- Portfolios dominated by TSLA clustered in the high-risk, high-return region
- Portfolios with higher MSFT and AAPL weights clustered in the low-risk, moderate-return region
- The **optimal portfolio** (highest Sharpe Ratio) was identified, representing the best risk-adjusted return available within this asset universe

CHAPTER 11 — VALUE AT RISK (VaR) ANALYSIS

11.1 Definition

Value at Risk (VaR) is a statistical measure that estimates the maximum expected loss of a portfolio over a specified time period, at a given confidence level, under normal market conditions.

Interpretation Example: A 1-day VaR of -2.3% at 95% confidence means: *"There is a 95% probability that the portfolio will not lose more than 2.3% of its value in a single trading day."* Equivalently, losses exceeding this threshold are expected only 5% of the time.

11.2 Methodology

The **Historical / Parametric VaR** method was used:

1. Computed the full historical distribution of daily portfolio log returns
2. Identified the **5th percentile** of this distribution as the VaR threshold
3. This threshold represents the worst expected daily loss at the 95% confidence level

Formula: `VaR_95 = Percentile(Portfolio Returns, 5)`

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11.3 Why VaR Matters

- VaR is a mandatory risk reporting metric under **Basel III** banking regulations
- Used by institutional investors, risk managers, and regulators worldwide
- Provides a single, comparable number to communicate risk exposure across different portfolios
- Enables risk budgeting — setting maximum acceptable loss thresholds before position adjustments are triggered

11.4 Limitations of VaR

- VaR does not describe what happens beyond the threshold (tail risk) — this is addressed by **Conditional VaR (CVaR)**, also known as Expected Shortfall
- Assumes normal market conditions; may underestimate risk during extreme events (black swan events)
- Historical VaR is backward-looking — it assumes future risk resembles historical risk

CHAPTER 12 — DASHBOARD DESIGN & VISUALIZATION

12.1 Design Philosophy

The Power BI dashboard was designed following three core principles:

- **Clarity:** Every visual has a single, clear purpose; no chart overloading
- **Interactivity:** Date slicers and asset filters allow users to dynamically explore the data
- **Business Readiness:** Layout and labeling follow professional financial reporting standards

12.2 Dashboard Components

Panel 1 — Stock Price Trends

- Multi-line chart showing adjusted closing prices for all 5 assets over the full historical period
- Interactive date range slicer for custom period selection
- Insight: Enables visual comparison of asset growth trajectories

Panel 2 — Returns & Rolling Volatility

- Dual-axis chart: bar chart for daily returns, line overlay for 30-day rolling volatility
- Highlights periods of market stress (e.g., 2022 bear market) vs. periods of calm
- Insight: Correlates return magnitude with volatility regime

Panel 3 — Correlation Heatmap

- Color-coded matrix (red = high positive correlation, blue = low/negative correlation)
- Provides instant visual insight into diversification opportunities
- Insight: Clearly shows TSLA's relatively lower correlation with other assets

Panel 4 — Monte Carlo Scatter Plot

- 10,000-point scatter plot: X-axis = Portfolio Risk (σ), Y-axis = Portfolio Return
- Color gradient by Sharpe Ratio — brighter colors indicate better risk-adjusted return
- Efficient frontier curve overlaid to highlight optimal portfolio zone
- Insight: Visualizes the complete risk-return landscape of all possible portfolios

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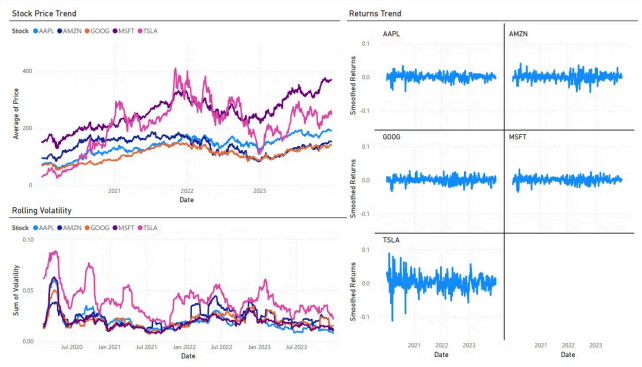
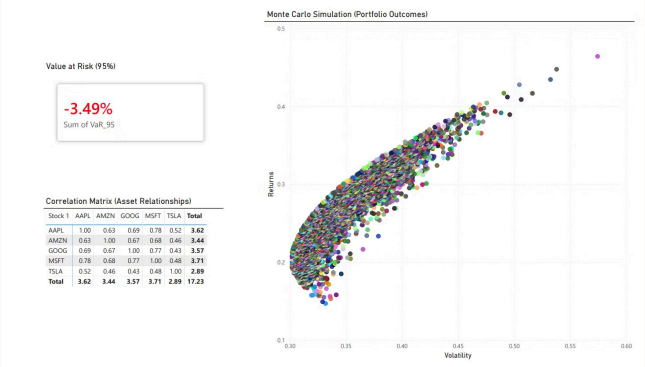
Panel 5 — VaR KPI Card

- Prominent KPI card displaying the 95% VaR figure prominently
- Supported by a return distribution histogram with the 5th percentile cutoff line marked in red
- Insight: Instant visibility into worst-case daily loss under normal conditions

12.3 BI Dashboard Preview



Portfolio Performance Overview



Risk & Simulation Analysis

CHAPTER 13 — KEY FINDINGS & INSIGHTS

13.1 Asset-Level Findings

- **TSLA** was the most volatile asset across the entire analysis period, exhibiting the widest return distribution and highest rolling volatility — confirming its high-risk, high-reward profile
- **MSFT** and **AAPL** demonstrated stable, relatively low volatility with strong, consistent positive returns — ideal core holdings for a conservative portfolio
- **AMZN** showed moderate volatility with periods of sharp drawdown followed by recovery — characteristic of growth-oriented assets
- **GOOG** displayed behavior most similar to MSFT, suggesting high substitutability between the two in a portfolio

13.2 Portfolio-Level Findings

- A diversified equal-weighted portfolio across all 5 assets produced **lower overall volatility** than any single asset held individually — empirically validating the diversification principle
- The correlation matrix confirmed that TSLA's inclusion meaningfully reduced overall portfolio correlation, improving diversification despite its higher individual volatility
- Monte Carlo simulation identified that the **optimal risk-adjusted portfolio** (highest Sharpe Ratio) was weighted toward MSFT and AAPL, with a moderate TSLA allocation for return enhancement

13.3 Risk Findings

- The 95% VaR estimate indicated a maximum expected daily portfolio loss under normal market conditions — providing a concrete risk boundary for position sizing decisions
- Periods of high rolling volatility (particularly early-to-mid 2022) corresponded with significant VaR increases, demonstrating that the risk model is responsive to changing market conditions

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CHAPTER 14 — LIMITATIONS & FUTURE SCOPE

14.1 Current Limitations

- **Data Scope:** Analysis limited to 5 U.S. equity assets; does not represent a globally diversified portfolio
- **No Live Streaming:** Data is fetched in batch mode; the system does not support real-time tick-by-tick updates
- **Static Risk-Free Rate:** A fixed risk-free rate was used for Sharpe Ratio calculations; in practice this changes daily
- **VaR Assumptions:** Historical VaR assumes future return distributions resemble past distributions — this fails during extreme market dislocations
- **No Transaction Costs:** Portfolio return calculations do not account for brokerage fees, taxes, or bid-ask spreads

14.2 Future Enhancements

- **Expand Asset Universe:** Include bonds, commodities, ETFs, and international equities for true multi-asset portfolio analysis
- **Conditional VaR (CVaR):** Implement Expected Shortfall to complement VaR and better capture tail risk
- **Machine Learning Integration:** Apply time-series forecasting models (LSTM, Prophet) to generate return predictions as Monte Carlo inputs
- **Live Data Streaming:** Integrate WebSocket-based real-time price feeds for intraday risk monitoring
- **Cloud Deployment:** Host the pipeline on AWS or Azure with scheduled daily data refreshes
- **Portfolio Optimization:** Implement mean-variance optimization (Markowitz framework) to generate the mathematically optimal portfolio weights
- **Stress Testing:** Model portfolio behavior under historical crisis scenarios (2008 GFC, 2020 COVID crash, 2022 bear market)

CHAPTER 15 — CONCLUSION

AlphaPulse successfully demonstrates how a comprehensive, end-to-end financial analytics and risk modeling system can be built using open-source technologies and publicly available financial data. By integrating data engineering, quantitative financial modeling, simulation techniques, and business intelligence visualization into a single cohesive pipeline, the project delivers a system that is both analytically rigorous and practically interpretable.

The system addresses real-world challenges in portfolio risk management — from data acquisition and cleaning through to advanced simulation and risk quantification — and presents results through a professional, interactive dashboard that meets business reporting standards.

Key accomplishments of the project include:

- A fully automated Python data pipeline from API ingestion to structured database storage
- Implementation of industry-standard financial metrics: log returns, rolling volatility, correlation, portfolio risk
- Monte Carlo simulation generating 10,000+ scenarios to map the complete risk-return landscape
- VaR estimation at 95% confidence — a regulatory-grade risk metric
- A five-panel interactive Power BI dashboard delivering consolidated, decision-ready insights

AlphaPulse stands as a strong demonstration of how data analytics and quantitative finance can be combined to build tools that are not merely academic exercises, but genuinely applicable to real investment decision-making.

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